Experiment 7: Part-of-Speech Tagging using NLTK in Python

1. Aim

To write a Python program using the NLTK library to perform part-of-speech (POS) tagging on a text.

2. Procedure

1. Import the `nltk` module in Python.
2. Define a sample sentence for part-of-speech tagging.
3. Use `nltk.word_tokenize()` to split the text into tokens.
4. Use `nltk.pos_tag()` to assign POS tags to each token.
5. Display the tokenized words and their corresponding POS tags.

3. Program

```
import nltk

text = "Python is a powerful programming language."

tokens = nltk.word_tokenize(text)
print("Tokens:", tokens)

tags = nltk.pos_tag(tokens)
print("POS Tags:", tags)
```

4. Output

```
Tokens: ['Python', 'is', 'a', 'powerful', 'programming', 'language', '.']
POS Tags: [('Python', 'NNP'), ('is', 'VBZ'), ('a', 'DT'), ('powerful',
'JJ'), ('programming', 'NN'), ('language', 'NN'), ('.', '.')]
```

5. Result

Thus, the Python program was successfully executed to perform part-of-speech tagging on a text using `nltk.pos_tag()`.